

Data Science undergraduate at the German International University (GPA 0.89, German scale $\approx$ 4.0/4.0; ranked 1st in cohort, Semester 1; full scholarship) who builds research systems end to end: preregistered ML experiments, large-scale data pipelines and shipped developer tools. Independent work spans **medical-LLM robustness**, **metric grounding of 3D foundation models**, **learned driving policies from racing telemetry**, and a **70k-line opportunity-intelligence engine** backed by 3,300+ tests.

EDUCATION

German International University (GIU) — *B.Sc. Informatics & Computer Science, Data Science major* Cairo · Expected 2028

- GPA 0.89 on the German scale (Excellent; $\approx$ 4.0/4.0 US) · Rank 1 in Semester 1 · Full tuition scholarship
- A+ in Mathematics I, Physics, Digital Logic, Programming I, Theoretical Computer Science, Computer Organization; A in Programming II, Computer Networks; $\sim$ 105% in Information Security
- Coursework: Data Science, Cryptography, Information Security, Computer Networks, Software Architecture, Software Testing, Mobile Security, Human-Computer Interaction

Nermien Ismail Schools (NIS) — *A Levels: A*A*A*B*

Cairo

RESEARCH PROJECTS

MedShift — *Robustness of Medical LLMs under Deployment Shift*

Independent research

Python · PyTorch · LoRA fine-tuning · Qwen3.5 (4B/9B) · MedGemma · GGUF/AWQ/GPTQ quantization · Vast.ai / Lightning / Modal GPUs

- Designed a benchmark for how clinical LLMs fail when deployed outside their training distribution: **GeoShift** (regional disease prevalence), **LanguageShift** (Arabic & code-switched patients) and **sensor shift** (pulse-oximetry and dermatology across skin tones).
- Built a **~4,000-example** training corpus across 8 shift families through a hash-verified pipeline with dual primary review, adversarial review and adjudication; auto-rejected invented labs, under-triage and test leakage.
- Fine-tuned quantized open models against a generic-data baseline under a preregistered analysis: **triage accuracy 58.1% $\rightarrow$ 71.9%** (+13.9 pp, Holm-corrected $p < 10^{-6}$, $n = 260$ blinded items), with **+40 pp** on GeoShift triage.
- Ran $\sim$ 1,000-generation inference campaigns with calibration, ablations and failure analysis.

Yardstick3D — *Metric Scale Recovery for 3D Foundation Models*

github.com/aequate/Yardstick3D

Python · NumPy · PyTorch · Depth Anything 3 · VGGT · COLMAP · SE(3) geometry

- Grounded **frozen** 3D foundation models (Depth Anything 3, VGGT) in metric scale using only sparse, noisy real-world cues (a GNSS displacement cue, plus Doppler-speed and altitude secondary checks), with no fine-tuning.
- Cut median SE(3)-aligned trajectory error on ADVIO from **8.06 $\rightarrow$ 1.42 m** and **7.38 $\rightarrow$ 1.68 m** (development), and **5.93 $\rightarrow$ 2.83 m / 5.90 $\rightarrow$ 2.25 m** on held-out sequences.
- Under a frozen protocol, grounded foundation models reached **1.23–1.42 m vs 4.12 m** for a COLMAP 4.2 structure-from-motion baseline at matched budget.
- Replicated prospectively across **two foundation models** (DA3, VGGT) and **two sites**, with pre-registered, held-out-flight validation.
- Oracle headroom analysis, claim ledger, adversarial review and **200+ tests**; evaluated on ADVIO, UrbanNav and MARS-LVIG (two sites).

F1 Autonomous Racing — *Telemetry $\rightarrow$ Behaviour Cloning $\rightarrow$ Control*

Independent research

Python · PyTorch · UDP / binary protocols · FastAPI · TypeScript / Vite · 712 tests

- Byte-exact real-time ingestion of EA F1 25/26 UDP telemetry at **60 Hz**: fuzz-tested parsers for both packet formats, raw-packet retention with hashing, frame-level sync and ML export.
- Trained an **event-gated residual behaviour-cloning policy**. On 16 laps recorded *after* the model was frozen: steering MAE **0.0096** (**2 $\times$ better** than persistence), **0/16** autoregressive divergences, brake-onset recall $\sim$ 0.81, throttle-lift recall $\sim$ 0.93; passed all **17/17** preregistered readiness gates.
- Built a 2D vehicle-dynamics simulator with a learned neural residual (system identification), counterfactual corner-time attribution and bounded vehicle-parameter optimisation (**12/12** and **13/13** gates).
- Shadow-mode real-time control loop with a safety limiter, watchdog and human override; FastAPI + TypeScript analysis workbench.
- **In progress**: validating the stack on real full-scale race-car data (Indy Autonomous Challenge RACECAR, AV-21): RTK-GNSS/IMU state fusion, parameters frozen before held-out laps, cross-team transfer.

Dance Sync CV — *Measuring Group Dance Synchrony from Video*

github.com/aequate/dance-sync-cv

Python · ONNX Runtime (CUDA) · YOLOX · RTMPose · multi-object tracking · signal processing

- Single 4K video $\rightarrow$ tiled detection, whole-body pose, offline tracking $\rightarrow$ sub-frame cross-correlation against group consensus motion.
- Recovers an injected 50 ms lag to within 20 ms; on 9 performances from the **IDL 2026 Seoul** series it reproduced the judges' final placement order.

ENGINEERING PROJECTS

Kymra / Opportunity Engine — Global Opportunity-Intelligence Platform

Founder & sole engineer

Python · PostgreSQL · FastAPI · Docker · async workers · TypeScript · React · WebGL · Playwright

- Built a ~70k-line Python system with **3,300+ tests** that discovers, verifies, deduplicates and ranks scholarships, internships, fellowships and research programmes, answering “*which am I actually eligible for, and why?*”
- End-to-end pipeline: search-led discovery frontier → acquisition → immutable snapshots → field-level extraction with provenance → canonicalisation and conflict records → authority verification → eligibility → ranking → semantic-change monitoring.
- **170+ source adapters** across North America, Europe, MENA and Asia; Arabic/multilingual extraction; lease-based scheduler and workers; hardened fetcher (SSRF protection, robots.txt, rate limits); evaluated on repeated **fresh blind hold-out** campaigns.
- Product frontend where every eligibility claim carries a quoted source “receipt”; WCAG AA contrast gate over 136 token pairs in two themes, **61 Playwright e2e tests** with axe on every route.

Git4Me — Open-Source Visual Git Client

github.com/aequate/git4me · v1.0

Rust · Tauri 2 · TypeScript · React 19

- Desktop Git client that shows history as an editable video-editor timeline: commits are clips, branches are tracks, HEAD is the playhead. Drag to cherry-pick, move or re-date commits.
- Rebasing plans are **replayed in memory** to predict conflicts before anything changes; every destructive action shows its risk and the equivalent Git command, and can be undone with Ctrl+Z.
- Hunk-level conflict resolver, GitHub sign-in / clone / publish, stashes, tags, submodules, GPG/SSH signing status, global search.

AUNI — University Student Network

Full stack

Next.js 16 · React 19 · TypeScript · PostgreSQL · Playwright

- Student platform launched for GIU: JWT auth, transactional migrations with advisory locks, persisted signature-based founding flow and a custom design system with animated page transitions.

SYSTEMS & VERIFICATION TOOLS

TensorWitness — SMT verification of quantized ML kernels

Python · Z3 · C++

Proves or refutes equivalence of int8 GEMM + requantization kernels with Z3 and emits minimal counterexamples. Caught **13/19** planted bugs vs **4/19** for random testing; found a rounding bug in ~18 ms that 5,000 random tests missed. Differentially checked against compiled gemmlowp C++ primitives.

ReplayLab — Deterministic replay for distributed-training failures

Python · multiprocessing

Records only the events needed to reproduce intermittent pipeline failures. **10/10** planted protocol bugs reproduced deterministically 5/5 times; causal reproducer synthesis shrinks failing runs to ~5% of their actions.

CipherCut — Parameter planner for encrypted (FHE) inference

Python · TenSEAL · CKKS

Searches legal CKKS parameters for privacy-preserving model inference. Beat the tutorial configuration on accuracy (**1.0 vs 0.9** agreement at equal latency) and ran **2.6× faster** than the library default.

COMPETITIONS

- ECPC 2026 qualification round: **ranked 67th**; team placed **top 3** among GIU teams
- Active on Codeforces and LeetCode; wrote a contest prep kit (C++ templates, pattern-recognition drills)
- GIU CTF 2025 participant (security capture-the-flag)

AWARDS

- **Full Tuition Scholarship**, German International University
- **Rank 1** in cohort, Semester 1
- Top-3 GIU team, ECPC 2026 qualifier

SKILLS

Languages	Python, C/C++, TypeScript/JavaScript, Rust, Java, C#, SQL
Machine learning	PyTorch, LoRA fine-tuning, quantization (GGUF/AWQ/GPTQ), LLM evaluation & calibration, behaviour cloning, pose estimation, ONNX Runtime, scikit-learn, pandas
3D vision	Depth estimation, camera geometry, SE(3) trajectory evaluation, structure-from-motion (COLMAP), 3D foundation models
Systems	PostgreSQL, SQLite, FastAPI, Docker, async workers & schedulers, web crawling, information retrieval, provenance, Z3/SMT, homomorphic encryption
Web & desktop	React, Next.js, Vite, Tauri, WebGL, Playwright, accessibility (WCAG)
Research practice	Preregistered protocols, frozen hold-outs, blinded adjudication, multiple-comparison correction, ablations, reproducible hashed artifacts
Spoken	Arabic (native), English (fluent), German (A2)